Flexibility markets for TSO and DSO networks

Jelle Meus
19 March 2025

KU LEUVEN

Jelle Meus

- Post-doctoral researcher at KU Leuven & EnergyVille
- Research areas: electricity market design, energy policy, energy system modeling
- Energy systems integration and modeling group (prof. Erik Delarue)
  - Provide quantitative tools and results to support an efficient operation of, and transition towards, a low-carbon energy system.
  - Tools: optimization, equilibrium modeling, agent-based modeling
  - Applications: assessment of energy policy and market design
    - E.g. Adequacy & capacity markets, (flow-based) market-coupling, grid tariffs, EU ETS, renewable support, …

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Flexibility markets for TSO and DSO networks

I. Introduction to electricity markets
   I. Actors and activities
   II. Market structure
II. Flexibility markets
   II. Balancing markets
   III. Implicit balancing
   IV. Congestion & flexibility markets

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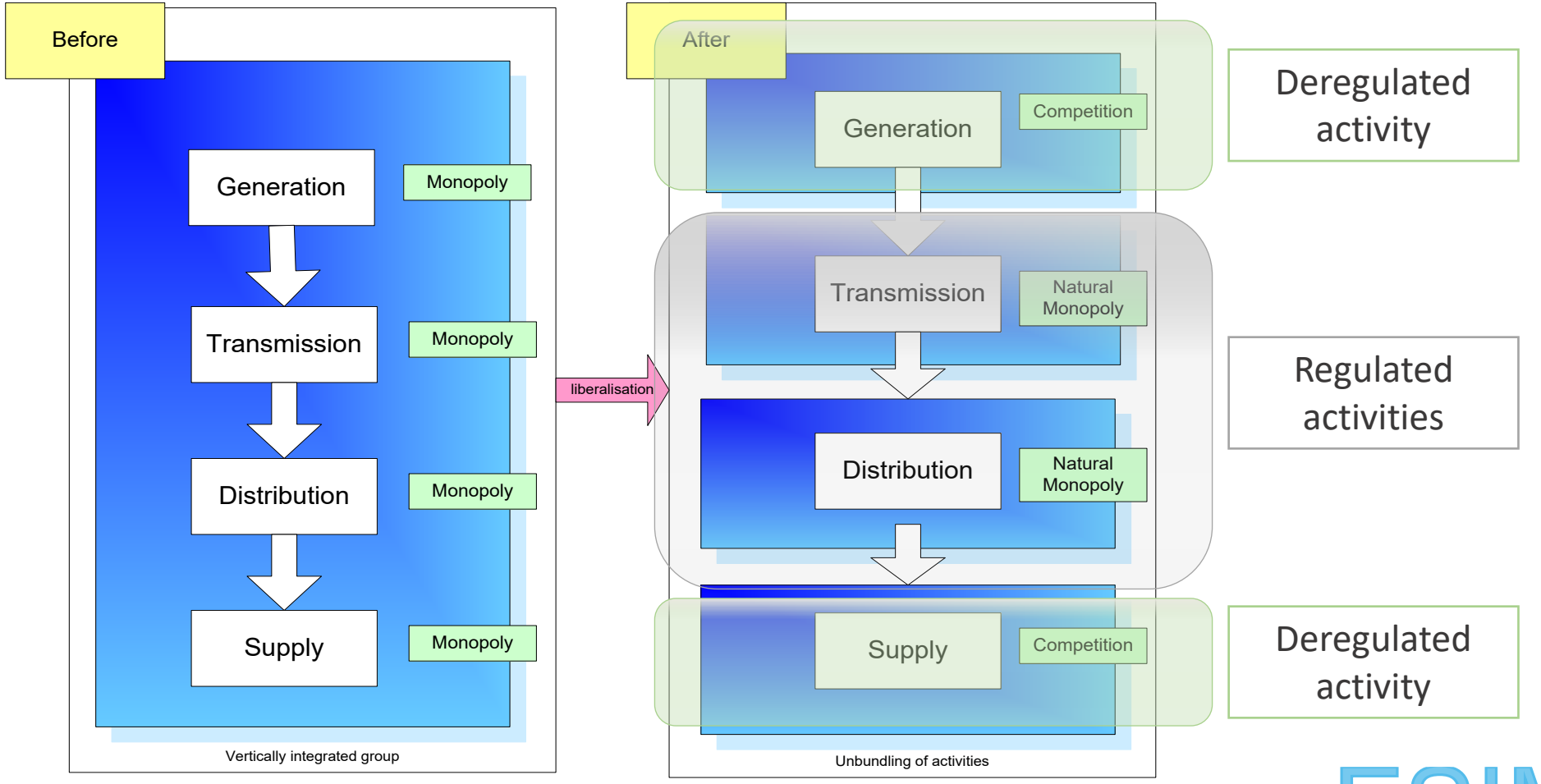
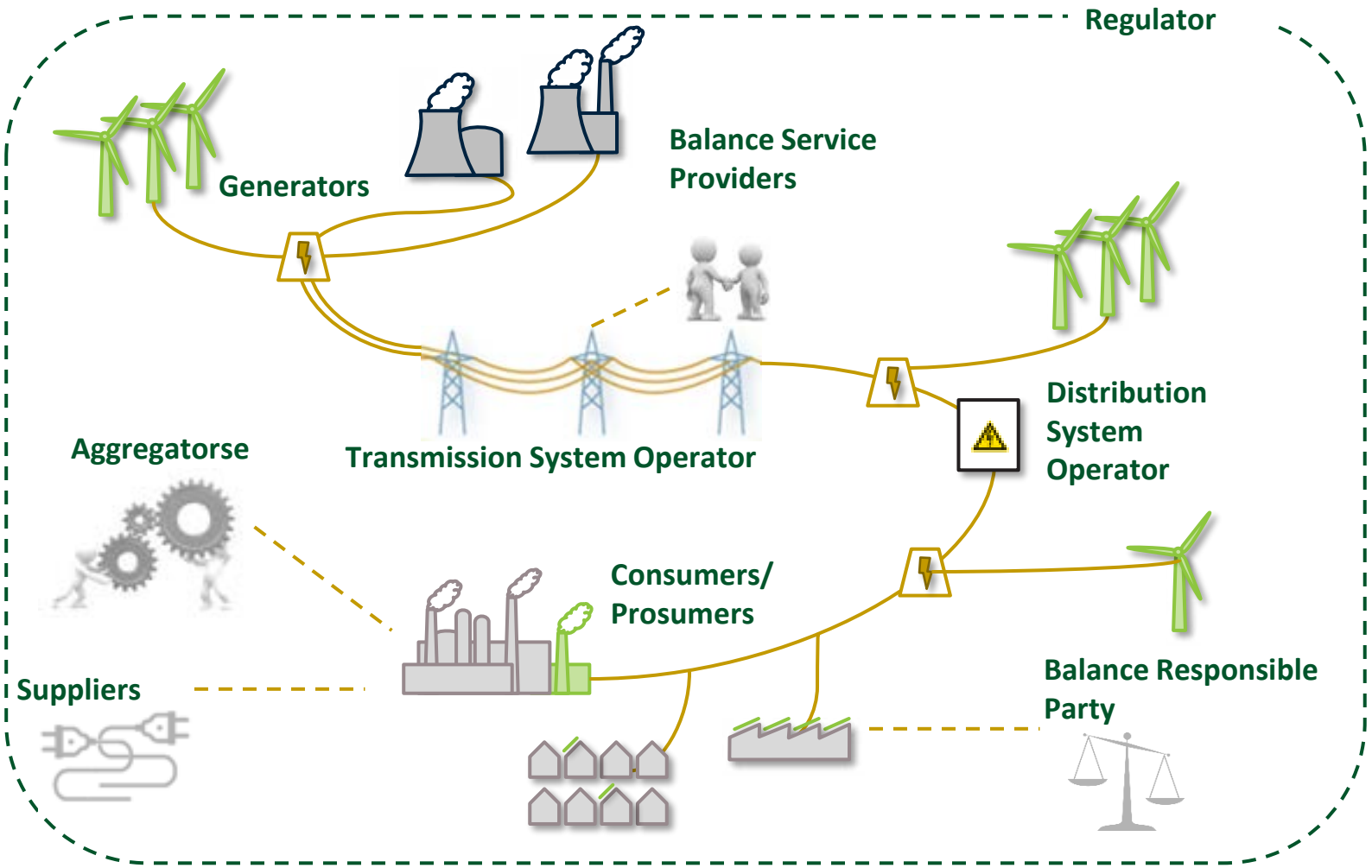
Activities of the power system: past and present



Actors and activities



The electricity grid:

Transmission lines BE: ~8 800 km

Distribution lines FL: ~134 000 km

Actors in electricity markets

Physical layer

Generators ⟷ **Transmission System Operator** ⟷ **Distribution System Operator** ⟷ **Consumer / Prosumer**

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Actors in electricity markets

Physical layer

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Market layer

Balance Service Providers

ENGIE
FLEXCITY by VEOLIA
EDF

Balance Responsible Parties

ENGIE
BASF The Chemical Company
TotalEnergies

Supplier

ENGIE
luminus
Eneco

Aggregator

FLEXCITY by VEOLIA
centrica Business Solutions
NEXT KRAFTWERKE

Traders

CITADEL
MERCURIA

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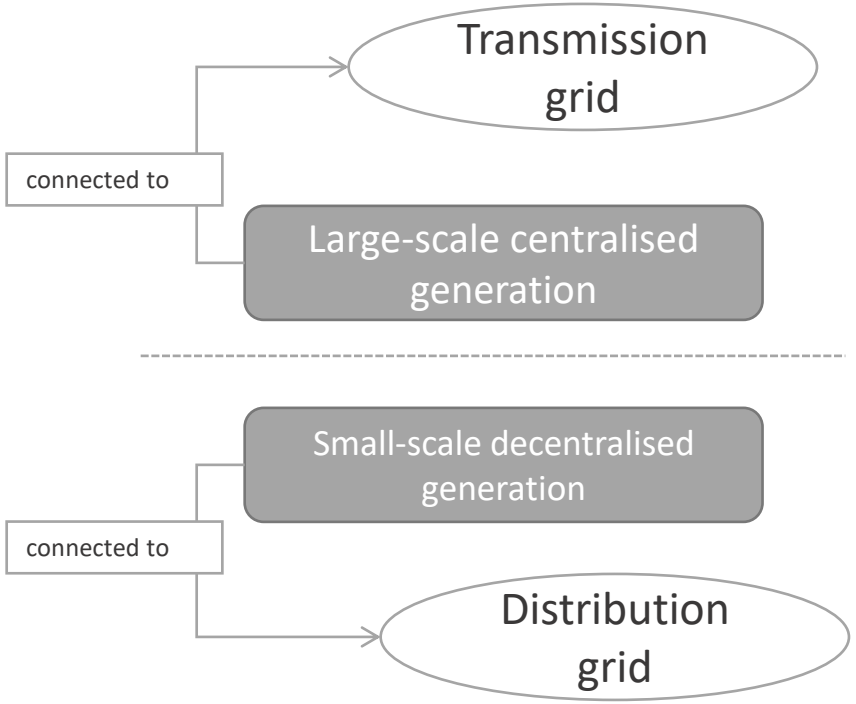
Actors and activities: physical layer

Generating companies

- Activities
  - Generate and sell electricity on wholesale level
  - Balance positions (BRPs)
  - Potentially flexibility provision (FSPs)
- Examples
  - Elia
  - EDF Luminus
  - ENECO
  - T-Power

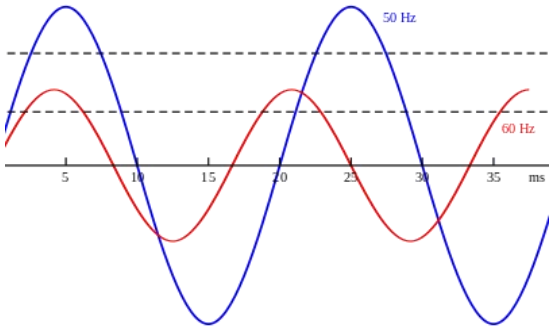


Actors and activities: physical layer

Transmission System Operator (TSO)

- High voltage transport of electricity
  - Transport longer distances (loss reductions)
  - International transport
- Feeder for
  - Distribution grid
  - Large industrial consumers (BASF, TotalEnergies, Umicore, Infrabel)
- Responsibilities
  - System stability
  - Grid investments
  - Interconnections
- Examples
  - Elia (Belgium)
  - RTE (France)
  - 50Hz (Germany)
  - TenneT (Netherlands)

Not uniquely defined:
Belgium: 70 – 380 kV
The Netherlands: 110 – 380 kV
United Kingdom: 132 – 400 kV



System stability

Grid investments

Interconnections

Actors and activities: physical layer

Distribution System Operator (DSO)

- Covers the "last mile" in the delivery of electricity
  - Linking the transmission system and electricity consumers at lower voltages
  - Medium and Low voltage transport of electricity
    - Downscale voltage transmission grid
    - Local distribution
  - Feed-in
    - Distribution-connected demand
    - Residential, industrial and tertiary sector

Not uniquely defined:
Belgium: up to 70 kV
Italy: up to 132 kV
France: up to 20 kV

- Responsibilities
  - Operation, maintenance & development of distribution grid
    - Construction & daily operation of distribution grid
    - Install new grid connections, adapt existing grid connections, install electricity meters
    - Solve problems & breakdowns in the distribution grid
    - Read electricity meters & manage consumption data
  - Public service obligations
    - E.g. promote rational use of energy, social supply, public lighting, …

Evolving role
Growth in distributed generation, digitalization, …

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Actors and activities: physical layer

Consumption

- Large consumers
  - Transmission (e.g. railways, BASF)
- Residential consumers, SMEs
  - Distribution
  - Usually passive, connection to electricity system is handled by supplier

Actors in electricity markets

Physical layer

Generators ⟷ Transmission System Operator ⟷ Distribution System Operator ⟷ Consumer / Prosumer

Market layer

Balance Service Providers

ENGIE
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CITADEL
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Actors and activities: market layer

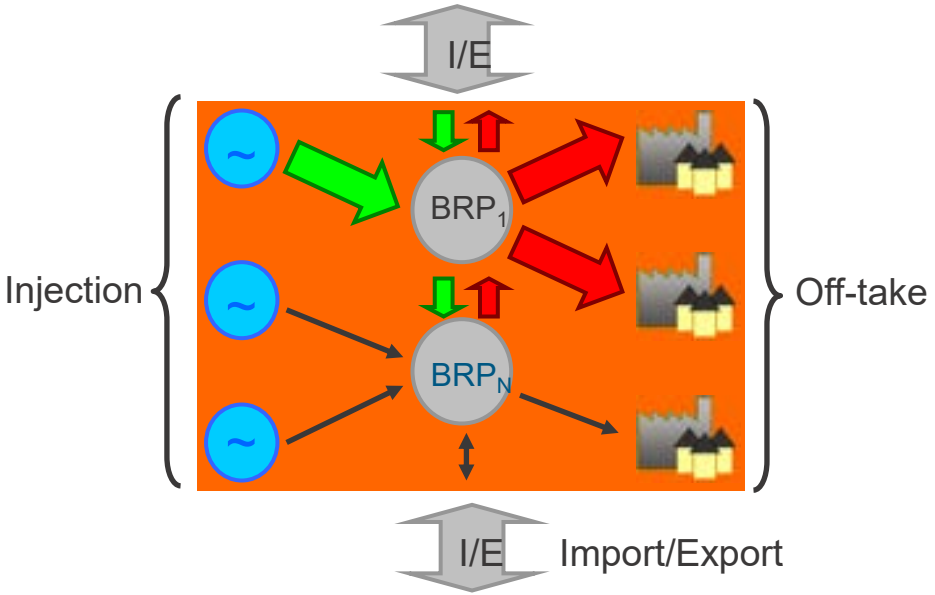
Suppliers

- Supply/sell energy to end-consumers
  - Buy electricity on wholesale market
  - Bilateral or market platform
- Charge electricity tariffs
  - Energy service, network tariffs, taxes
  - Fixed versus variable tariffs
- Balance portfolio or assign BRP
  - Estimate demand
- Offer public service obligations
  - e.g. invoicing, information, customer services, …
- Examples
  - Engie, Luminus, TotalEnergies, Eneco, …

Actors and activities: market layer

Balance responsible party (BRP)

- Contributes to maintaining system balance
- Each BRP holds a portfolio of injection and offtake
- Every significant producer and consumers are mandated to have a contract with a BRP (or take up the role of BRP themselves)
  - Generators
  - Industrial offtakers
  - Suppliers
- The BRP's portfolio needs to be balanced
  - Adjusting production and consumption of assets within portfolio
  - Trading with other BRP's on the wholesale markets
- Unbalanced portfolios can cause system imbalances
  - TSO is ultimately responsible for the system's stability and will take actions to restore balance
  - TSO charges their incurred cost to unbalanced BRPs



Actors and activities: market layer

Balance service providers (BSP)

- The TSO intervenes to address system imbalances
  - Increase production or decrease consumption when there is a shortage of generation
  - Decrease production or increase consumption when there is excess generation
  - Potentially also activate these reserves cross-border
- TSOs cannot own generating assets
- TSOs instead procure reserves from **balance service provides**
  - In advance, the TSO selects and remunerates BSPs to remain available (regardless of activation)
  - In case of a system imbalance, the TSO activates BSPs and remunerates them for their activation cost

- Who can be a BSP?
  - Power plants
  - Large demand facilities
  - Batteries/storage technologies
  - Aggregators
  - …

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Actors and activities: market layer

Aggregator

- Mediator between consumers and generators, markets & other power system actors
- Gathers ("aggregates") the flexibility of consumers (and generators) to "build" Active-Demand (AD) services
- Typically does not own any physical assets

Flexibility markets for TSO and DSO networks

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II. Flexibility markets

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The European electricity market design

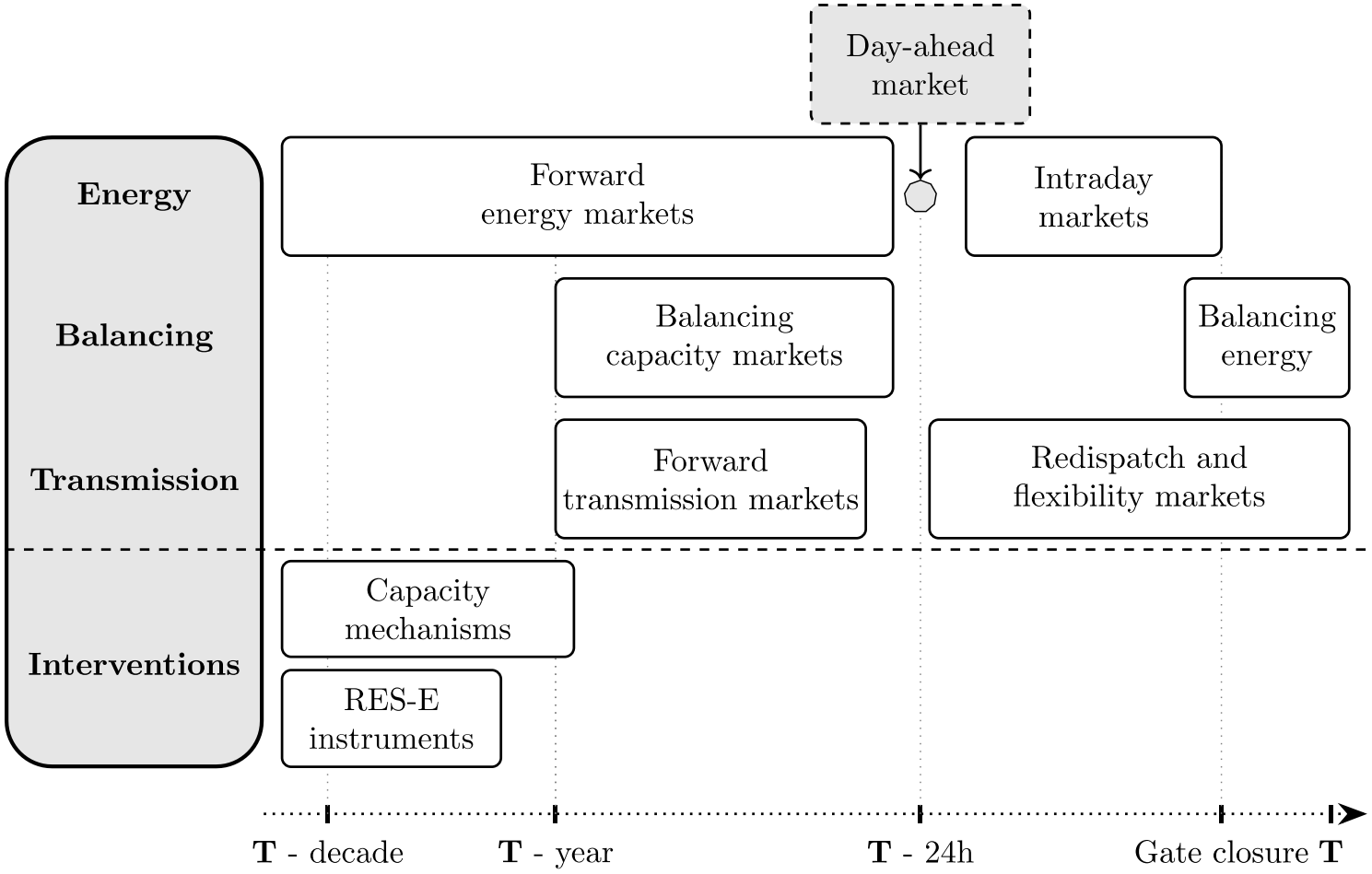
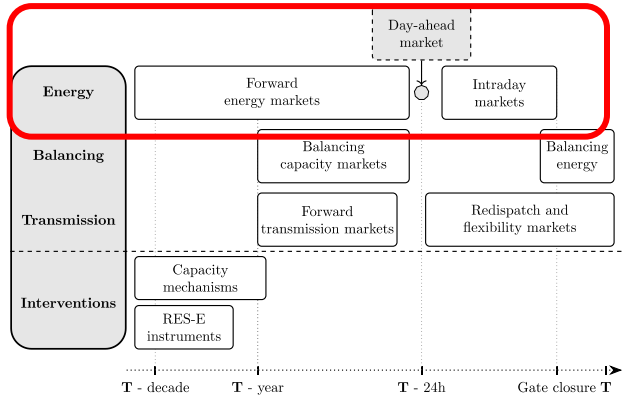
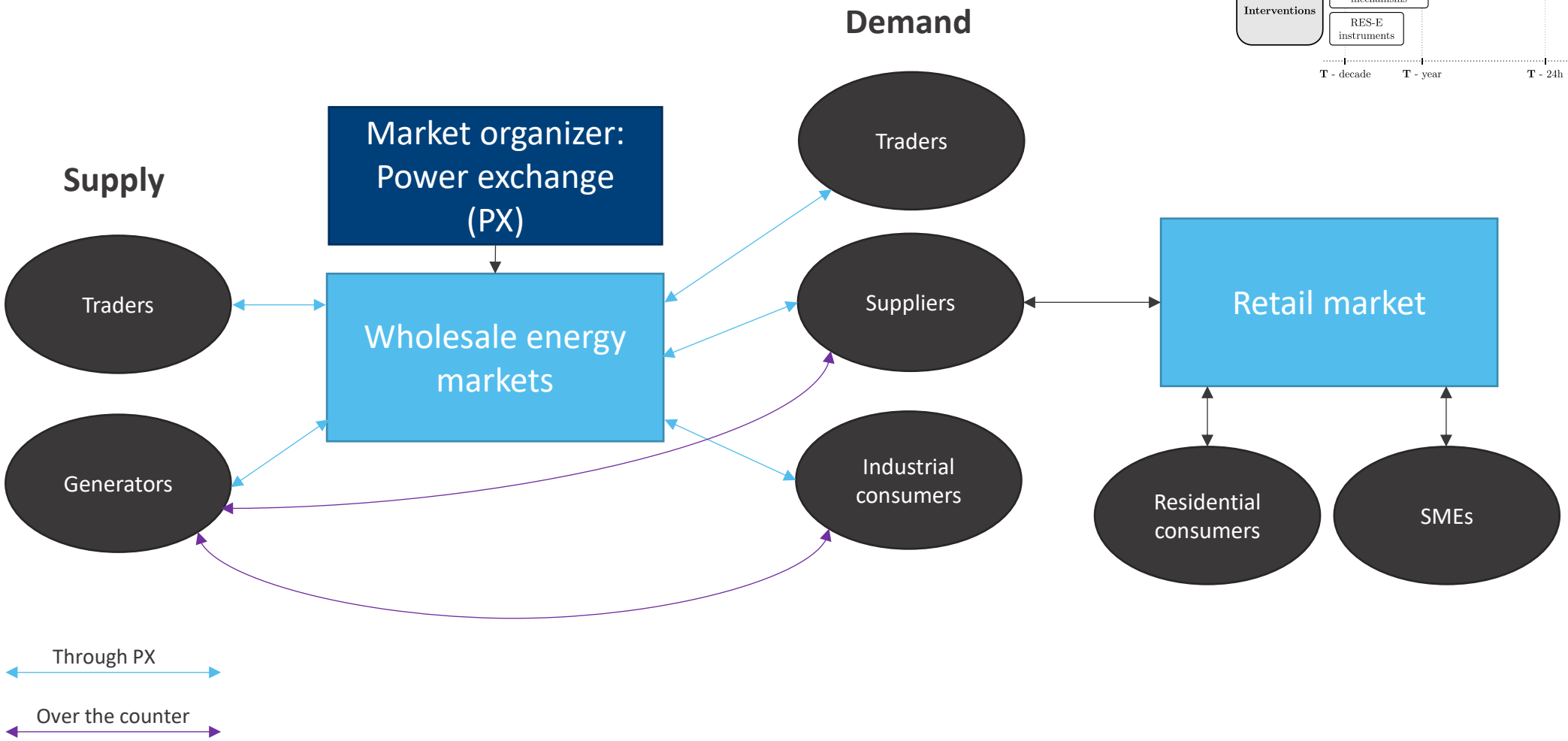


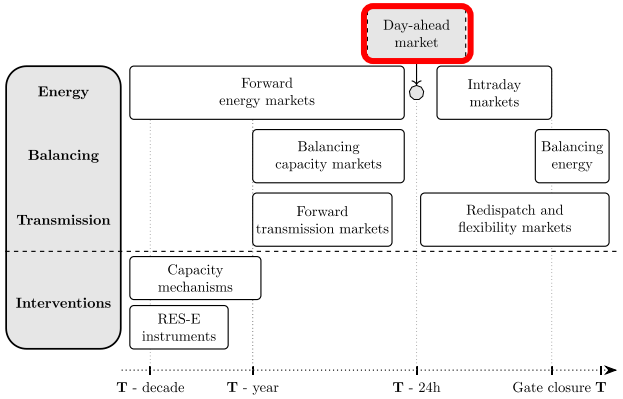
Figure adapted from: Meeus, L. (2020). The evolution of electricity markets in Europe. *Edward Elgar Publishing.*

Energy trading: participants

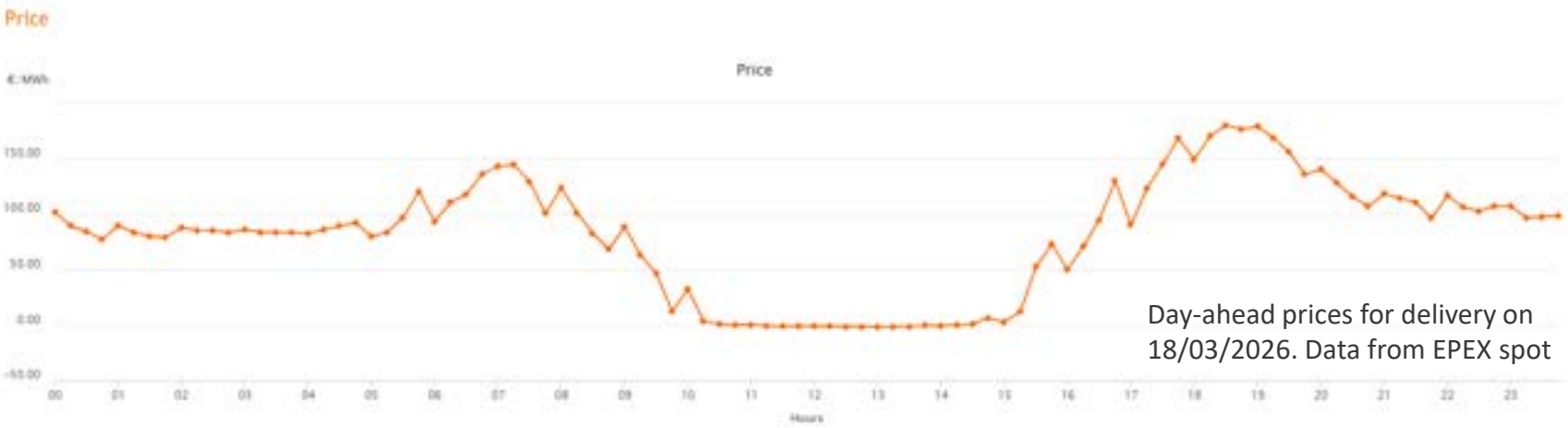




The day-ahead market



- ‘Reference market’ for electrical energy
- Determines market outcomes for all 15-minute intervals of next day
  - Organised 365 times per year
  - Produces 96 electricity prices and 96 sets of production and consumption schedules per day (one per quarter-hour)
    - Note that it a simultaneous clearing for all 15-minute intervals within a day, given interdependencies.
    - Note that the quarter-hourly resolution is recently introduced (October 2025). Before that, it was hourly.
- Market participants submit their price-quantity bids by noon
- Market organizer (NEMO) clears market and publishes results about an hour later



Day-ahead prices for delivery on 18/03/2026. Data from EPEX spot

Pricing electricity

1) Market participants submit price-quantity bids to the power exchange

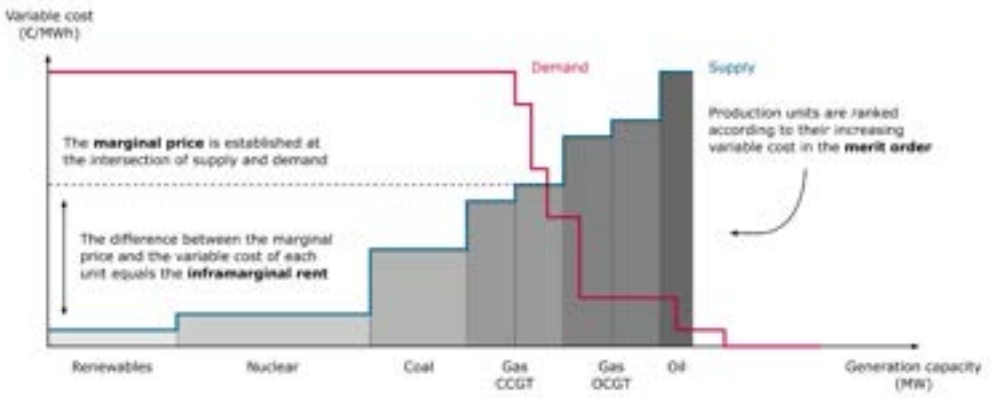
- Generators offer the quantity they can produce and the minimum price they require to produce it
- Consumers offer the quantity they want to consume and the maximum price they are willing to pay

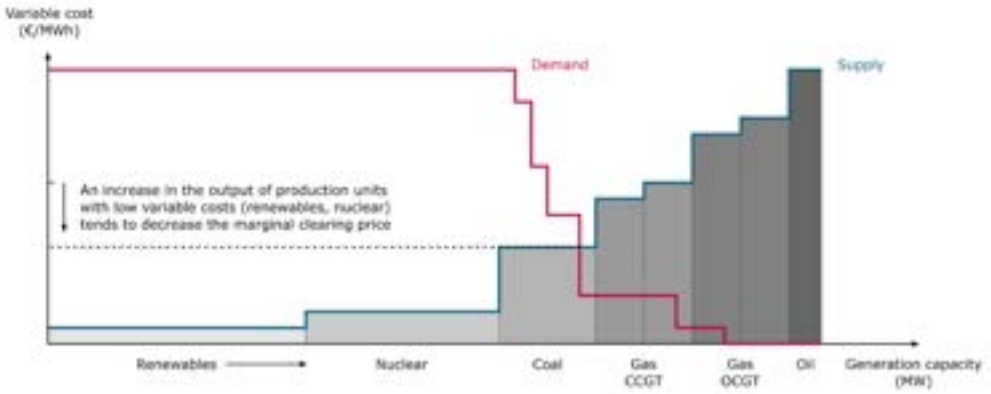
2) The bids are translated into demand and supply functions

3) Price is set by demand-supply intersection

4) The market outcome

- Generators with bids lower or equal than market price are generating
- Consumers with bids higher or equal than market price are consuming





Single day-ahead coupling (SDAC)

Multi-regional coupling (MRC)

- Day-ahead market is cleared for multiple market zones (countries) simultaneously
  → Increases price convergence and welfare
- Covers 19 countries, about 85% of power consumption
- **Implicit** price coupling

SDAC in a nutshell

- Market participants submit bids to NEMOs
- TSOs compute available cross-border transmission capacities and share these as optimisation constraints
- EUPHEMIA is the market coupling algorithm (ran by NEMOs)
  - Maximises welfare (i.e. matching of bids) subject to transmission constraints.
  - Produces prices and net positions per bidding zone.

https://fsr.eui.eu/electricity-markets-in-the-eu/

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Intraday electricity markets

Purpose

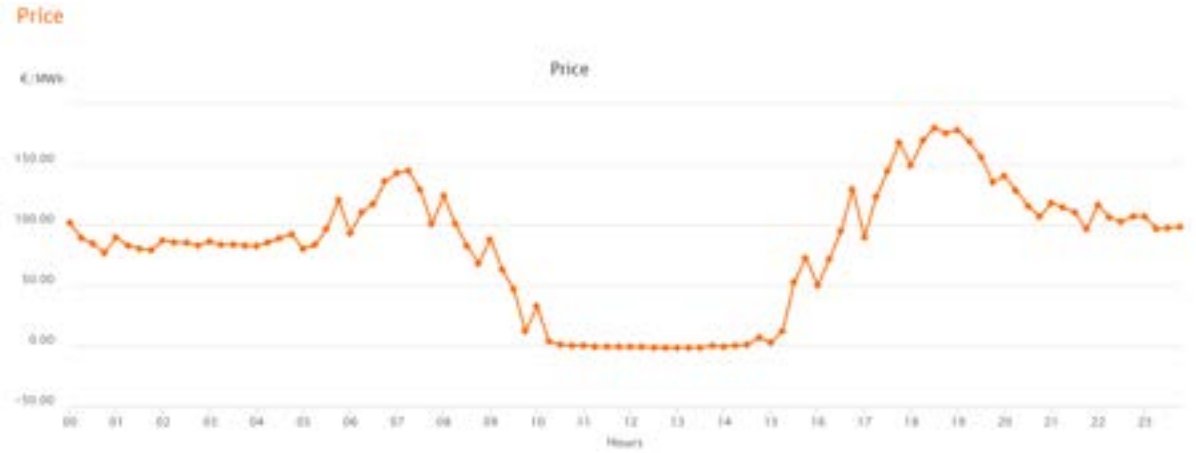
- Allow BRPs to more easily balance their portfolios
- DA bids still subject to uncertainty (wind and solar output, demand, power plant availability, etc.)
- What if power plant breaks after day-ahead commitment?
  - Producer could make up for deficit by activating another (potentially expensive) asset
  - Producer could also buy additional electricity on intraday market from other parties

Format: both continuous trading and auction-trading

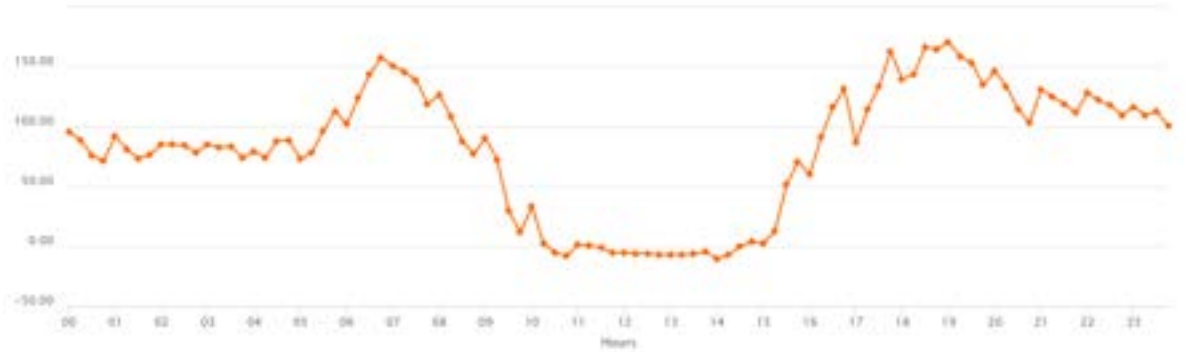
- Auction-trading:
  - Similar to day-ahead markets
  - Organised three times per day (15:00 D-1, 22:00 D-1, 10:00 D)
  - 15-min resolution
- Continuous market:
  - Market participants place bids continuously
  - Trade is executed as soon as two bids can be matched
  - No unique prices for same product
  - 15-minute resolution

Continuous intraday prices for delivery on 18/03/2026. Data from EPEX spot

Day-ahead and intraday market outcomes 18/03



Day-ahead prices



Intraday 15:00

Intraday 22:00

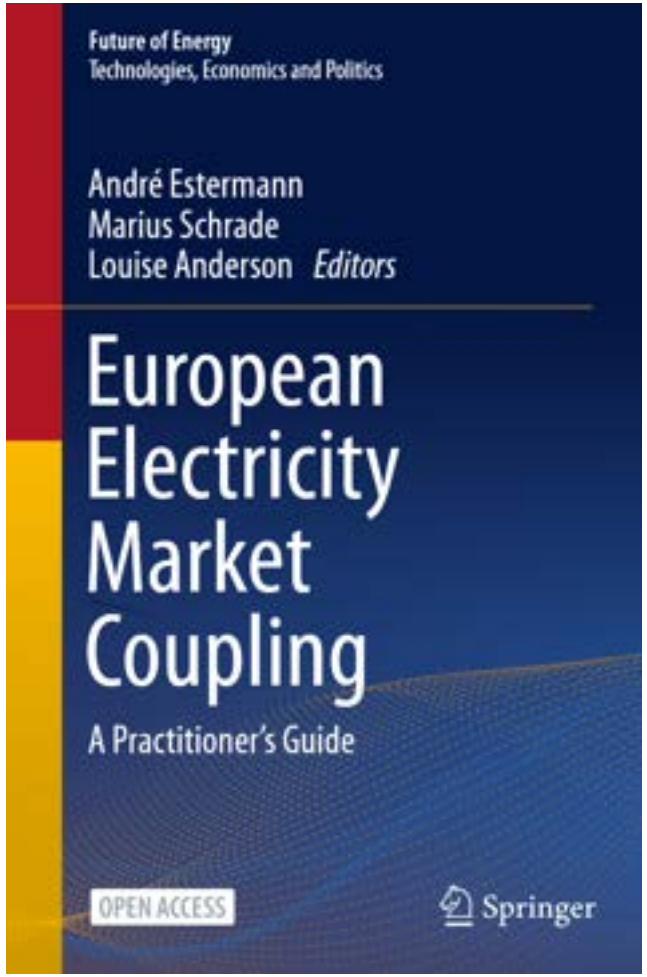
Single intraday coupling (SIDC)

Auctions

- Same principle as day-ahead market coupling.
- Implicit coupling
  - TSOs submit capacity to market platform
  - Market participants submit bids to market platform
  - EUPHEMIA clears market given transmission limitations

Continuous trading (XBID)

- Continuous trading uses a so-called limited order book
  - Capacity management module (CMM) tracks and updates state of cross-border capacity
  - Shared order book module (SOB) aggregates all given orders into a consolidated order book
  - Local trading Solution (LTS) is what traders see: bids and offers in SOB filtered by available transmission capacity

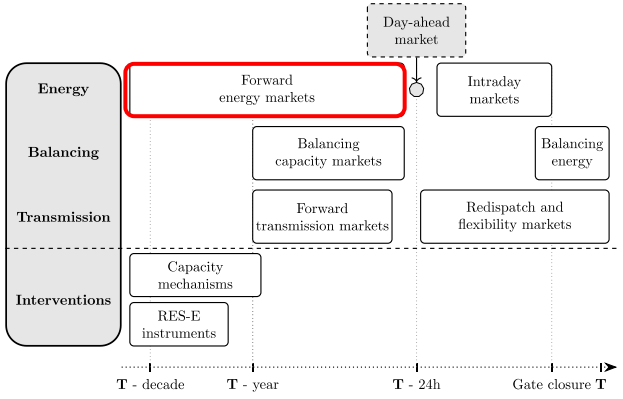


Future electricity markets

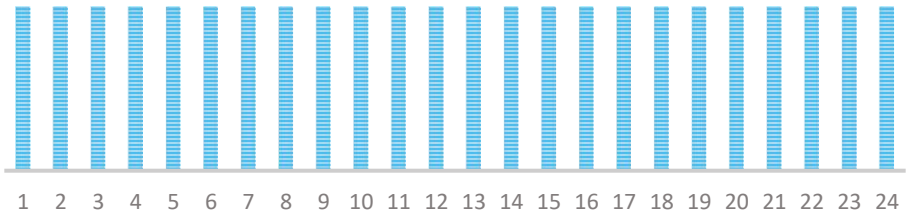
- Allow parties to hedge their positions for financial certainty
- Allows market participants to trade **profiles**
- Products vary by:
  - Duration: yearly, quarterly, monthly, sometimes weekly and daily
  - Profile: base, peak
- Liquidity strongly decreases for lead times of 3 years and more
- Difference forward/future markets:
  - Forward: bilateral (e.g. PPAs)
  - Future: standardized products on exchange

Belgian baseload futures (18/03/2026)

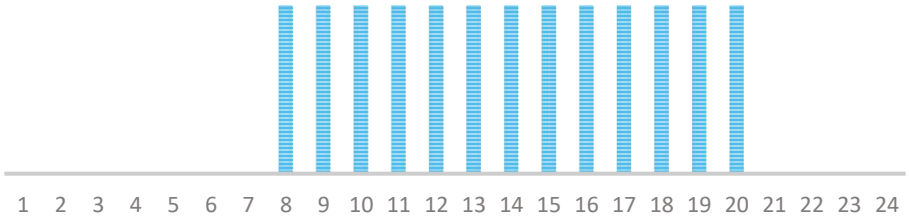
Yearly durations		Quarterly durations		Monthly durations	
Cal-27	85.3	2/26	93.7	April	98.4
Cal-28	70.4	3/26	98.2	May	87.7
Cal-29	67.6	4/26	107.6	June	95.0
Cal-30	-	1/27	103.8	July	-
Cal-31	-	2/27	72.6	August	-



BASE PROFILE



PEAK PROFILE



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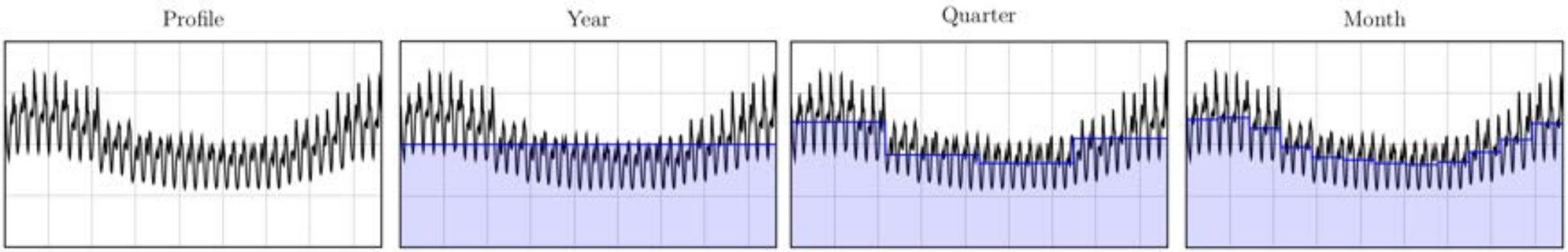
Future electricity markets

Example: suppliers or large consumers procuring energy



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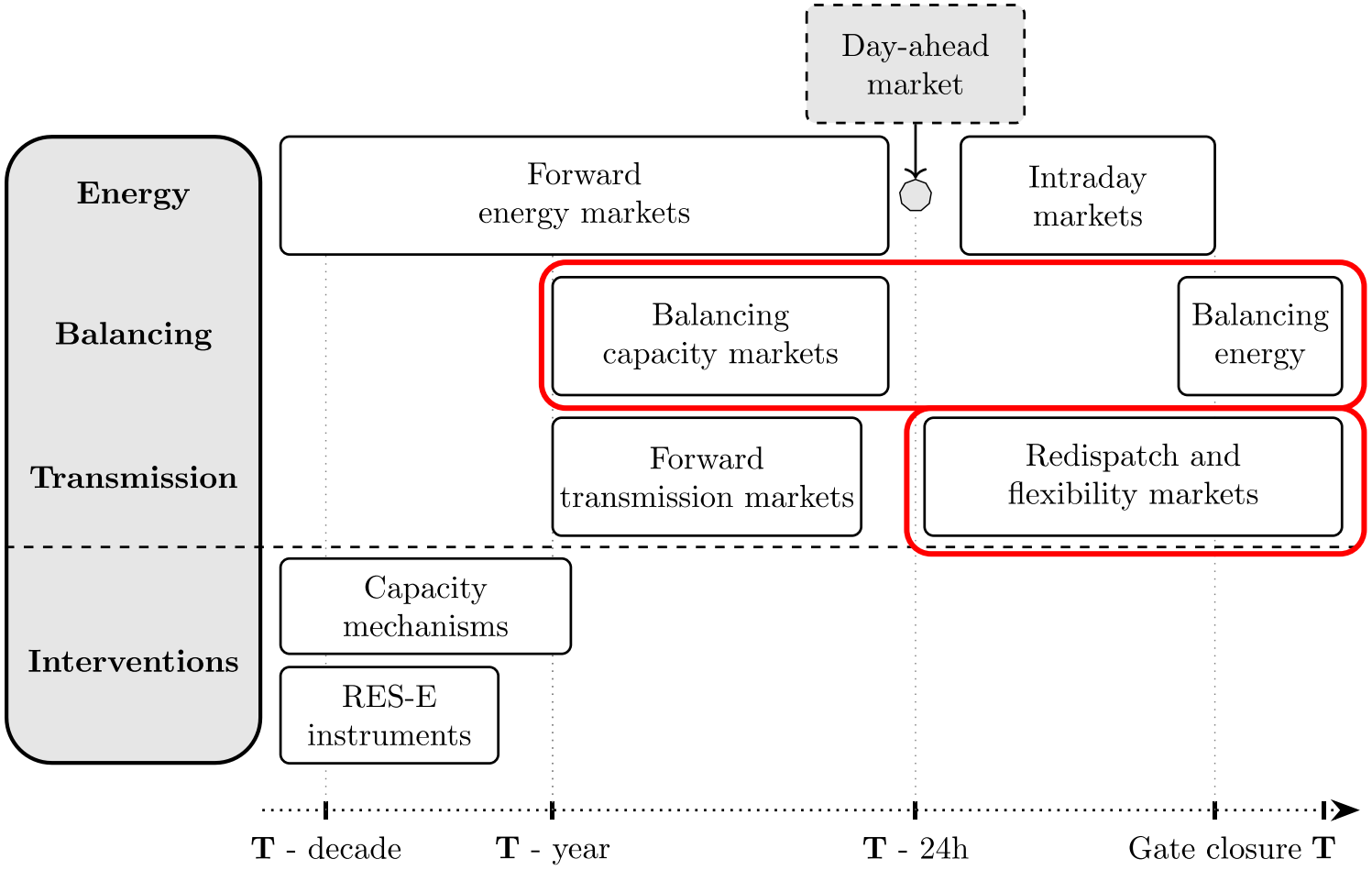
The European electricity market design



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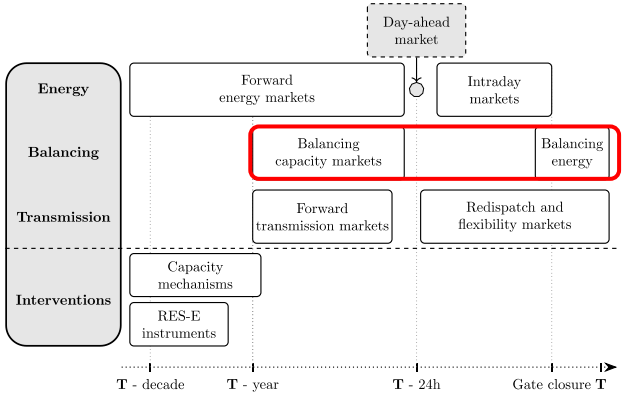
Balancing markets: the issue



System imbalance in BE on 18/03/2026

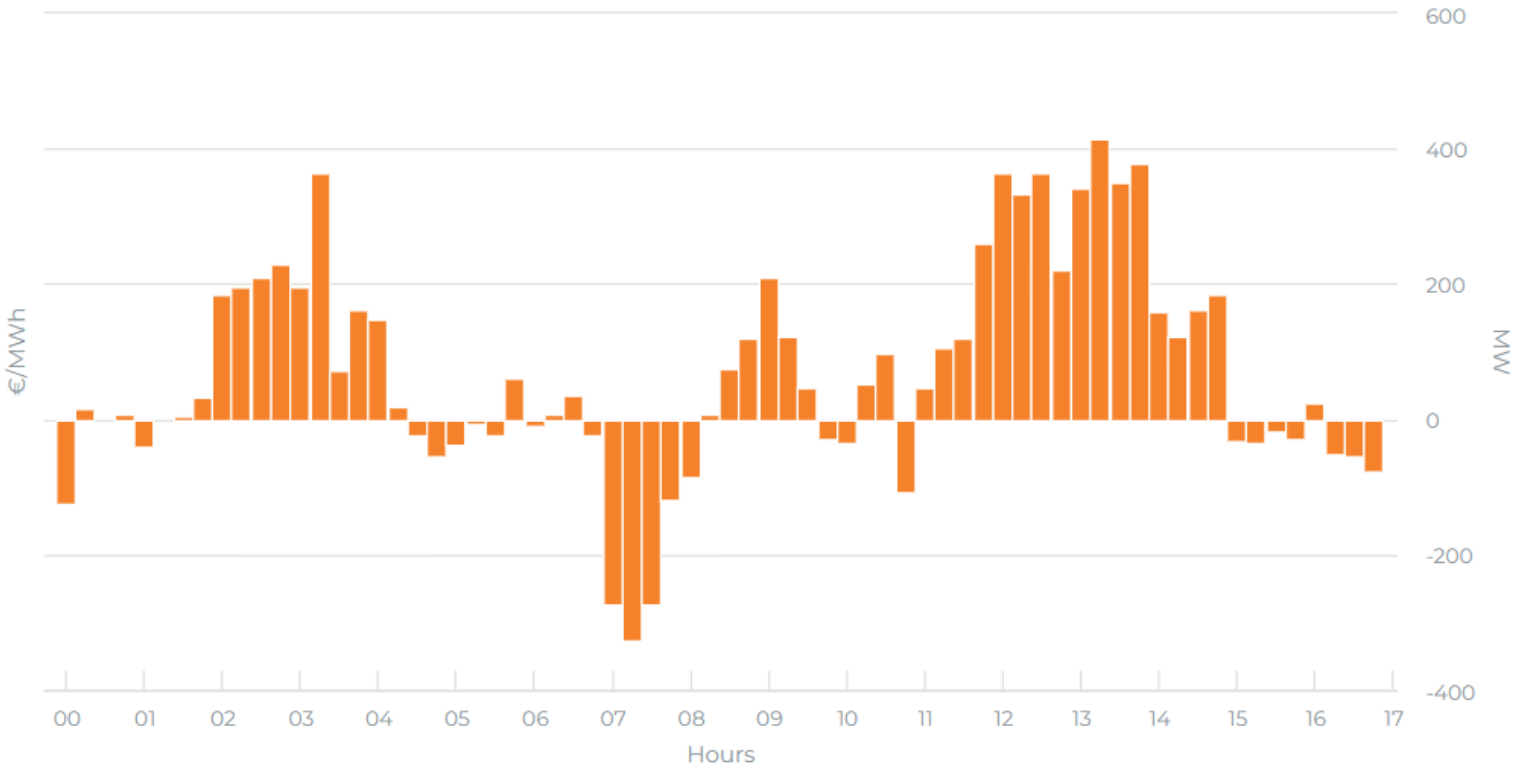


Figure taken from Elia

Balancing markets: the issue

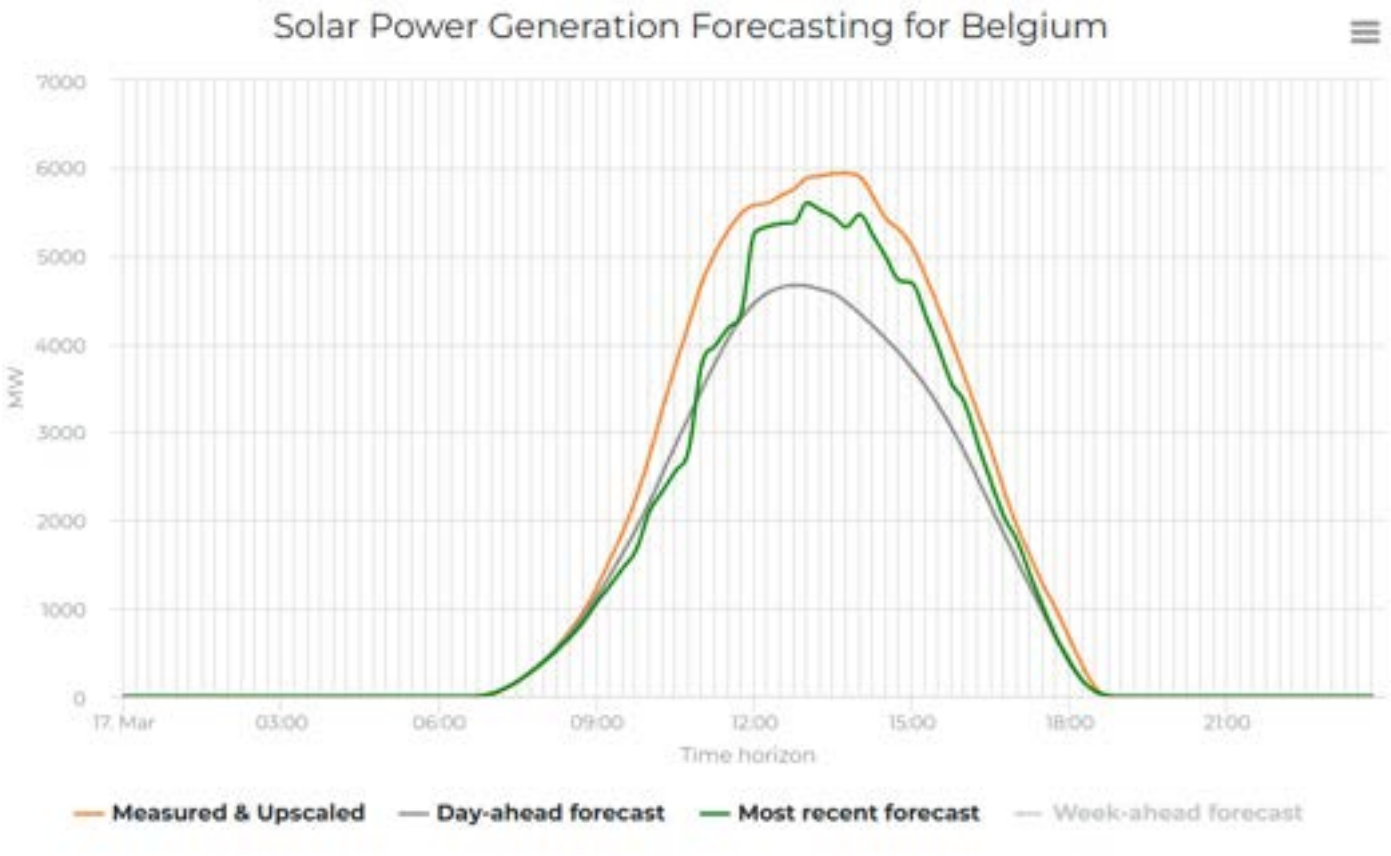
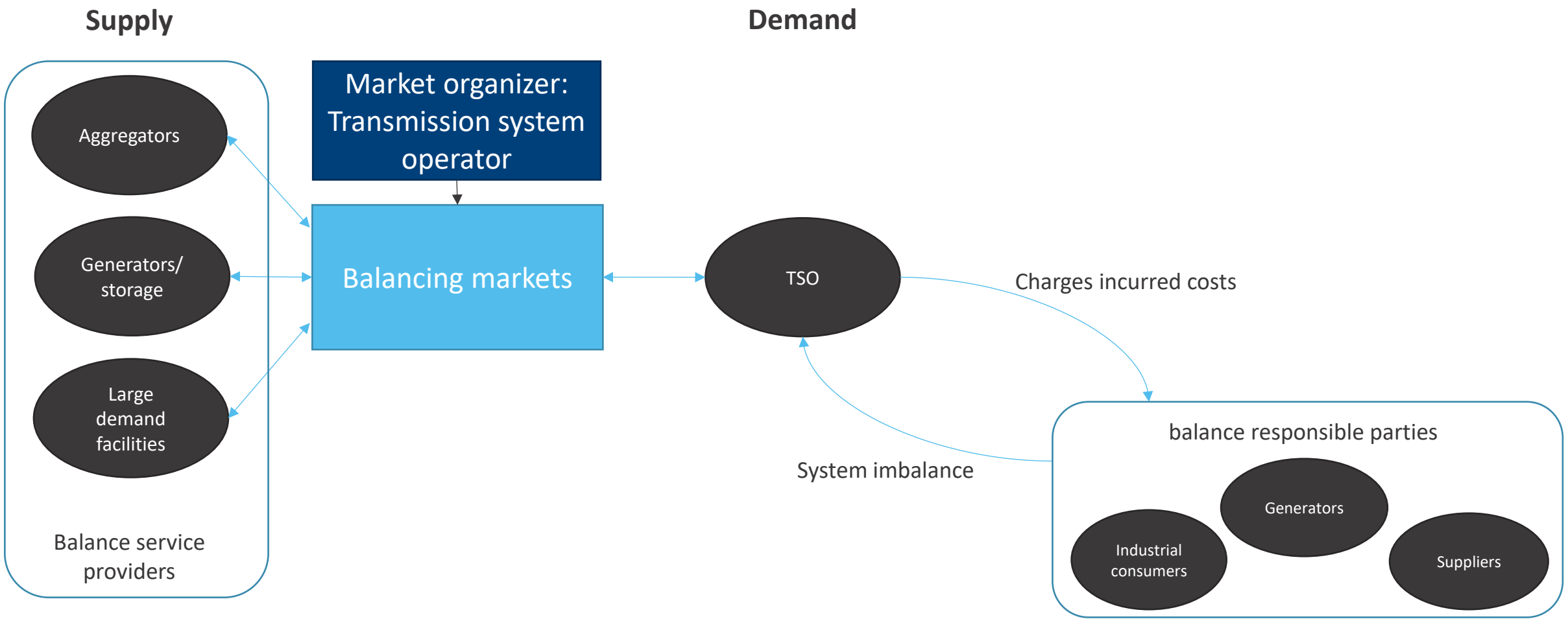
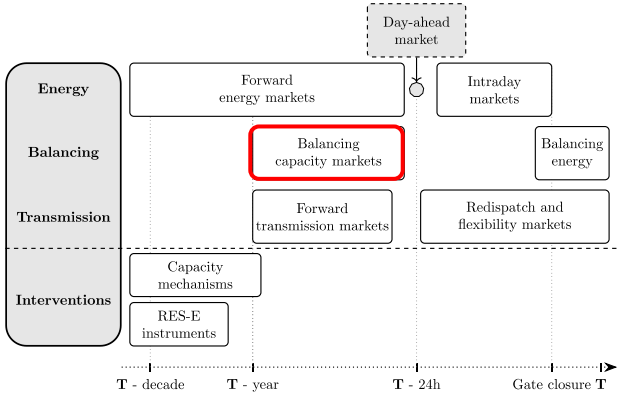


Figure taken from Elia

Balancing markets: participants



Balancing capacity markets



1. The TSO determines reserve capacity ahead of time
   - Both upward and downward capacity
   - Can vary over time

2. The TSO organises periodic auctions to procure reserve capacity from **balance service providers**
   - Before day-ahead clearing
   - BSPs offer price-quantity bids to TSOs
   - TSOs select cheapest ones to fulfill reserve requirement
   - Price is in €/MW, capacity or availability remuneration
   - Note that this payment is granted regardless of activation

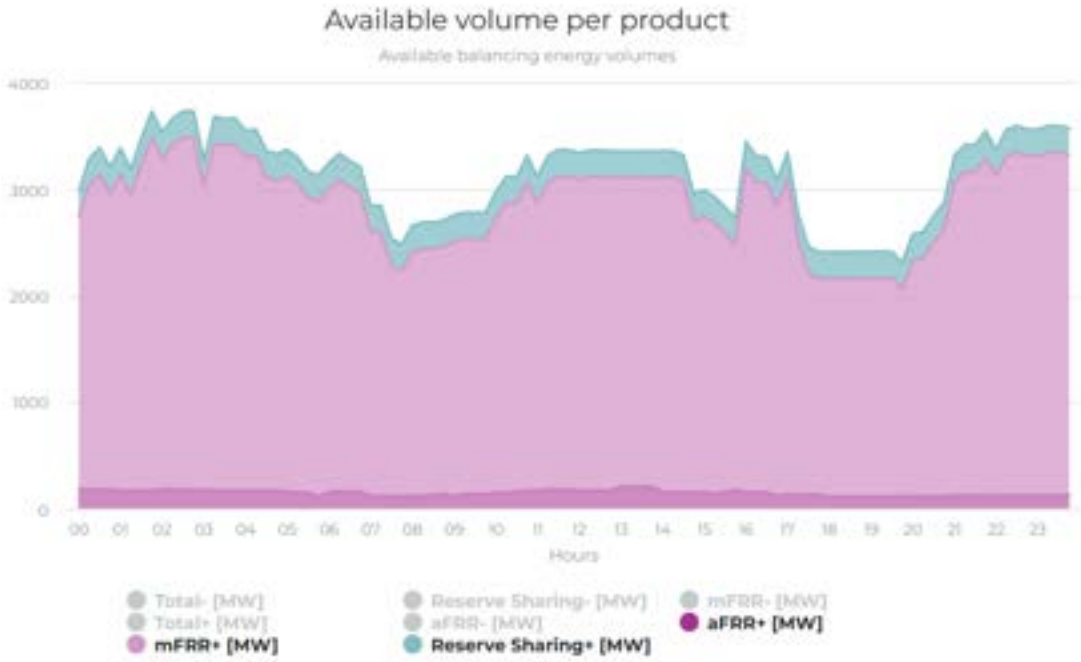


Figure taken from Elia

Balancing markets: products

Frequency containment reserves (FCR)

- Contain frequency deviations
- Activation time: 30 seconds
- Continuous activation
- ~90 MW
- Revenues: ~70k/MW/y

Automatic Frequency Restoration Reserve (aFRR):

- Restore frequency
- Activation time: 5 minutes
- Continuous activation
- ~120 MW
- Revenues: ~120k/MW/y

Manual Frequency Restoration Reserve (mFRR):

- Restore huge imbalances
- Activation time: 12.5 minutes
- Limited activation (~20 per year)
- ~1000 MW
- Revenues: ~50k/MW/y

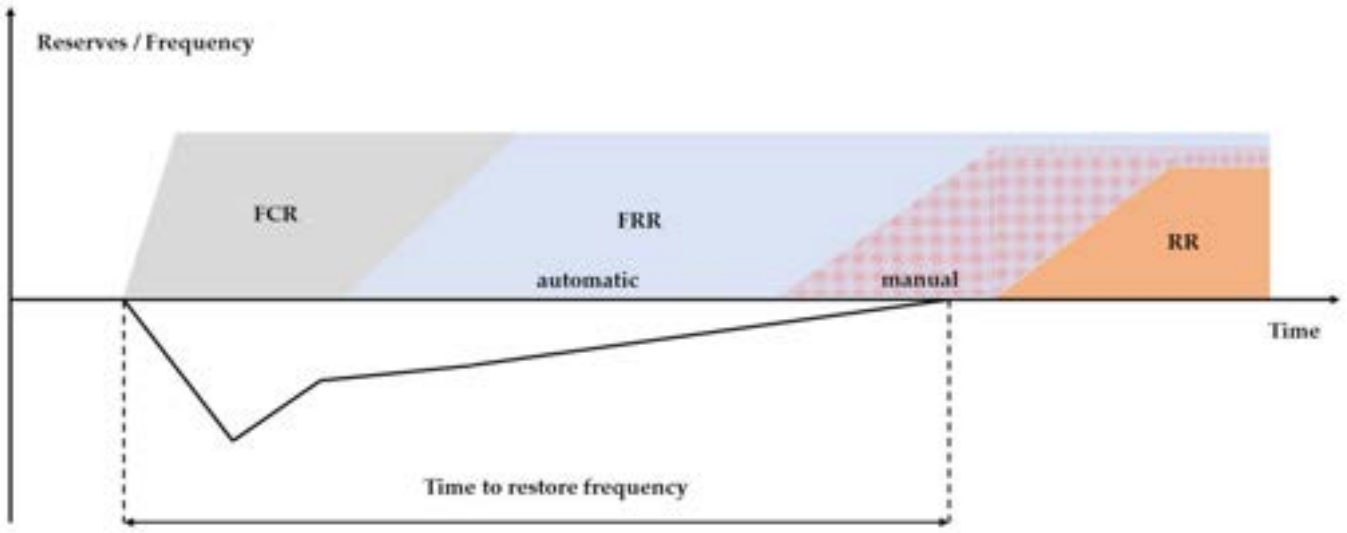


Figure from: Roumkos C, Biskas PN, Marneris I. Manual Frequency Restoration Reserve Activation Clearing Model. *Energies*. 2021; 14(18):5793. https://doi.org/10.3390/en14185793

Balancing markets: timing

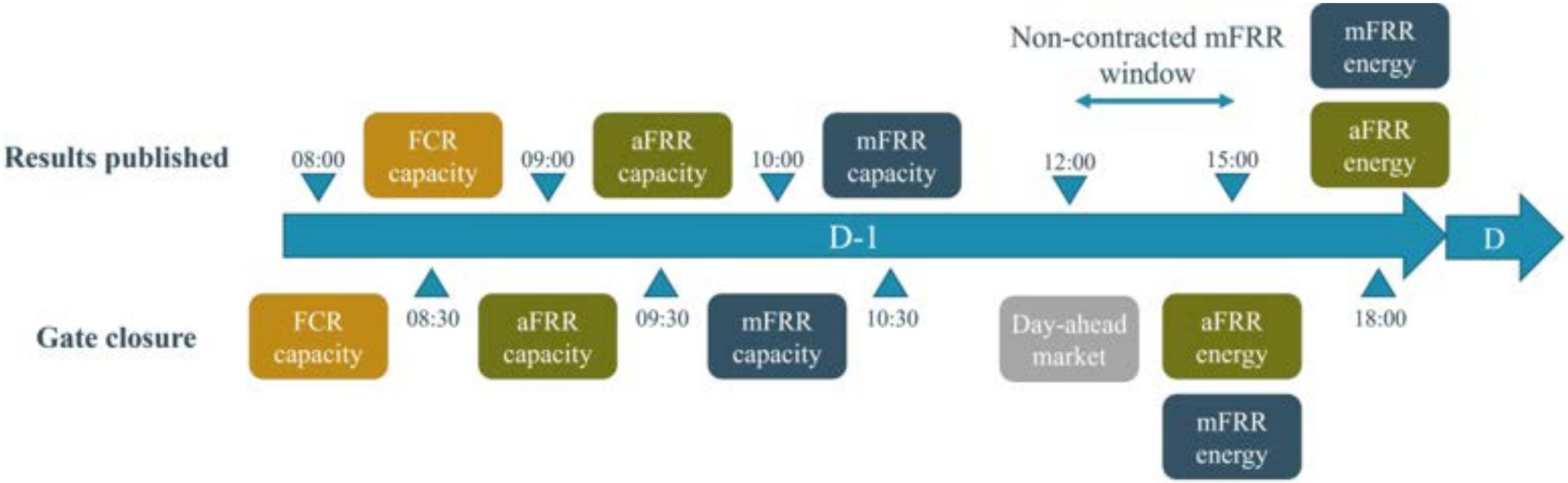
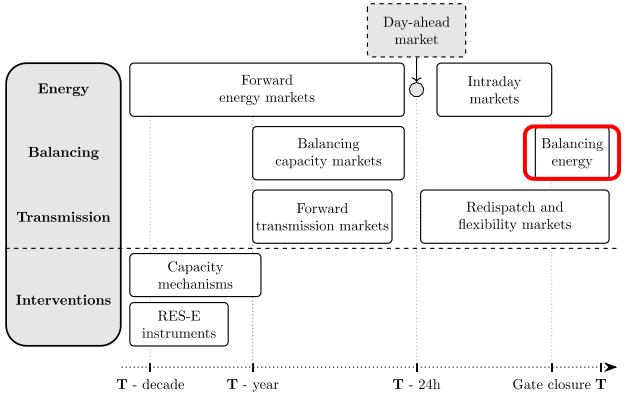


Figure taken from R. Smets, “Participation of energy storage systems in short-term electricity markets: Exploring the interaction between optimization and machine learning,” PhD dissertation, KU Leuven, Leuven, Belgium, 2024.

Balancing energy markets



- When a system imbalance occurs, TSO sends control signal to BSPs
  - Following the merit-order principle: cheapest reserve providers get activated first (within reserve category)
  - BSPs procured in balancing capacity market, or free bids
- BSP activates and receives activation remuneration (in €/MWh), corresponding to their bid
- **Imbalance price**
  - the marginal cost of the most expensive BSP required to restore the system imbalance
  - To be paid by BRPs if shortage, received if excess
    - But prices can be negative!

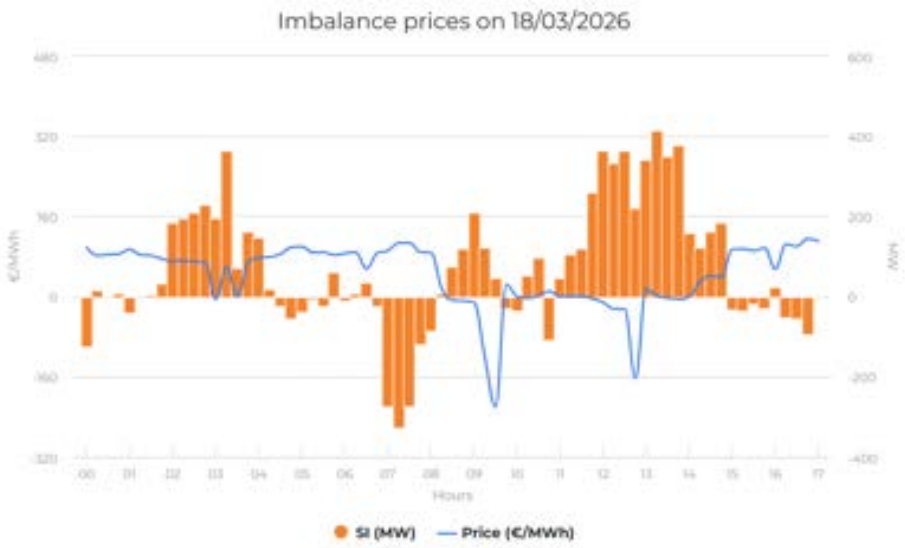


Figure taken from Elia

Cross-border balancing activation

- IGCC (2010):
  - International Grid Control Cooperation
  - Elia Belgium joined in 2012
- PICASSO (2022):
  - Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation
  - Elia Belgium joined in 2024
- MARI (2022)
  - Manually Activated Reserves Initiative
  - Elia Belgium joined in 2025

PICASSO

MARI

https://www.entsoe.eu/network_codes

Cross-border reserve markets

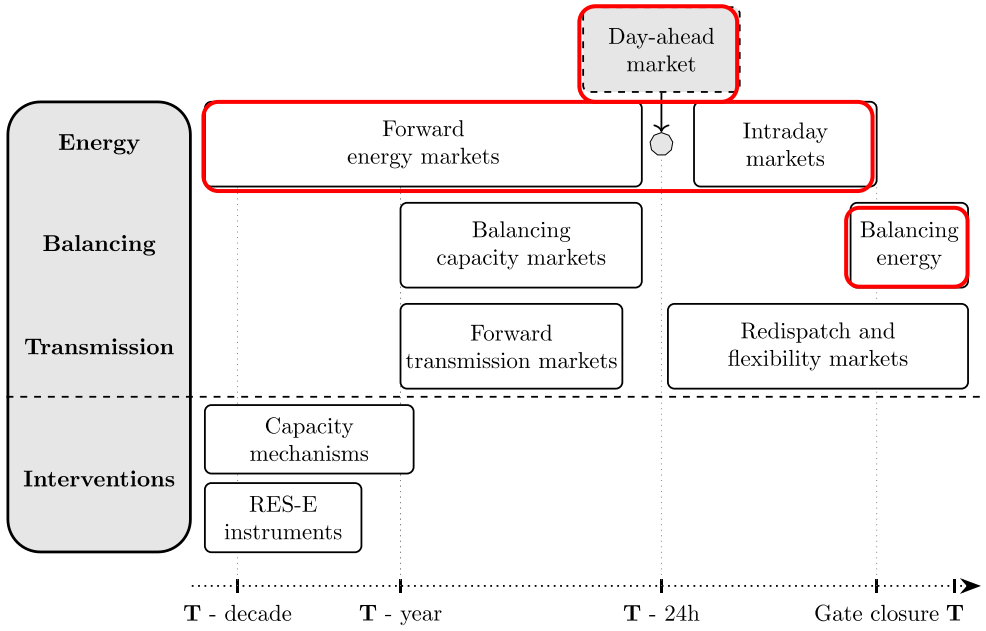


Table 1: Degrees of cooperation in cross-border balancing between TSO zones.

PROCUREMENT of reserve capacity	ACTIVATION of balancing energy
To meet the reserve requirements resulting from reserve dimensioning	To meet real-time imbalances resulting from forecast errors and unforeseen events
Autarky: no cross-border cooperation **Exchange**: procure reserves in other zones **Sharing**: multiple zones take into account the same reserves	**Autarky**: no cross-border cooperation **Imbalance netting**: avoid counteracting activation **Exchange**: activate reserves in other zones

F. M. Baldursson, E. Lazarczyk, M. Ovaere, and S. Proost, "Cross-Border Exchange and Sharing of Generation Reserve Capacity," *The Energy Journal*, vol. 39, no. 4, pp. 57–84, Jul. 2018,

Flexibility markets for TSO and DSO networks

I. Introduction to electricity markets

II. Flexibility markets

  II. Balancing markets

  III. Implicit balancing

  IV. Congestion & flexibility markets

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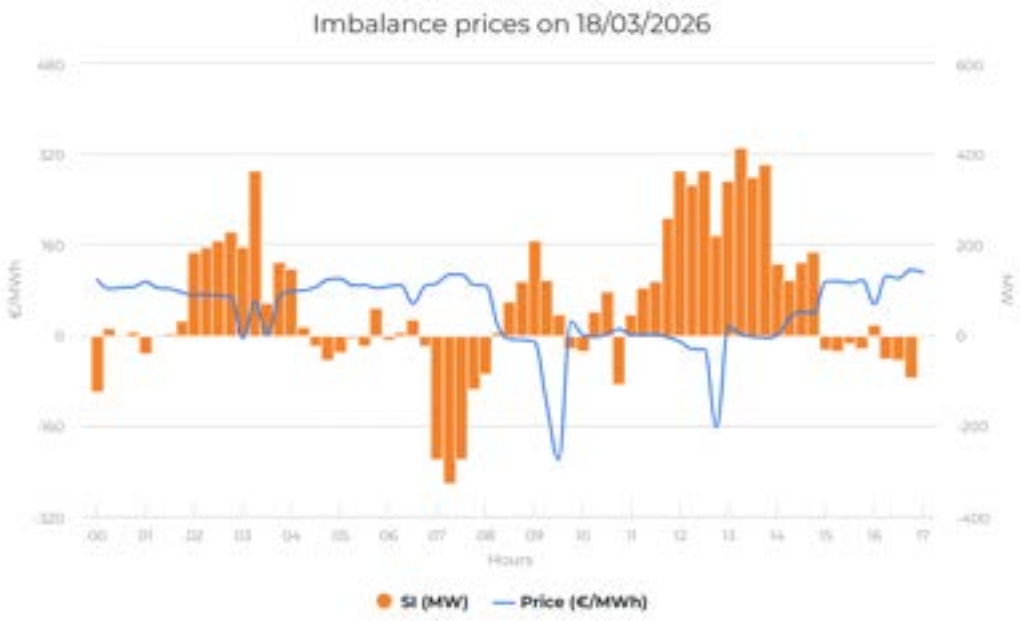
Implicit balancing

- BRPs actively taking imbalance positions to gain imbalance price
  - BRPs in a long position (more electricity produced than consumed) receive the imbalance price
  - BRPs in a short position pay the imbalance price
  - (Note that the imbalance price can become negative)

- Efficiency strongly depends on design of this imbalance (or real-time) price
  - Elia, for example, is currently moving from ‘imbalance payments’ to a ‘real-time price’
  - Tennet (NL), on the other hand, were concerned about system stability and delayed the publishing of reserve activation and imbalance prices in December 2024.

- (Sometimes called passive balancing)



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Redispatch and flexibility markets

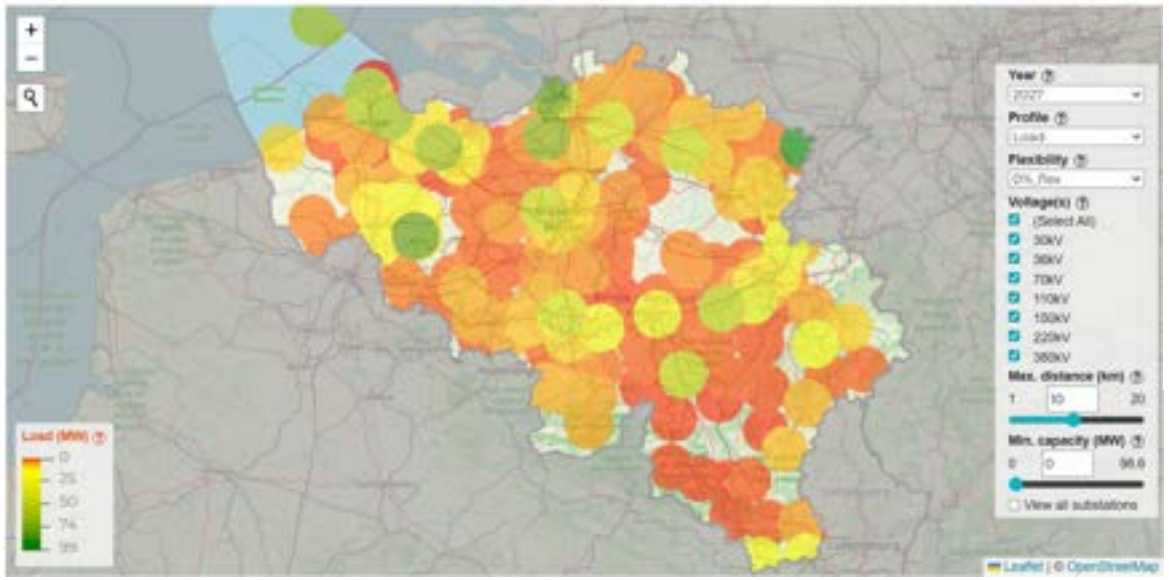
- Congestion markets: aim is to solve congestion on TSO level
- Flexibility markets: aim is to solve congestion on DSO level

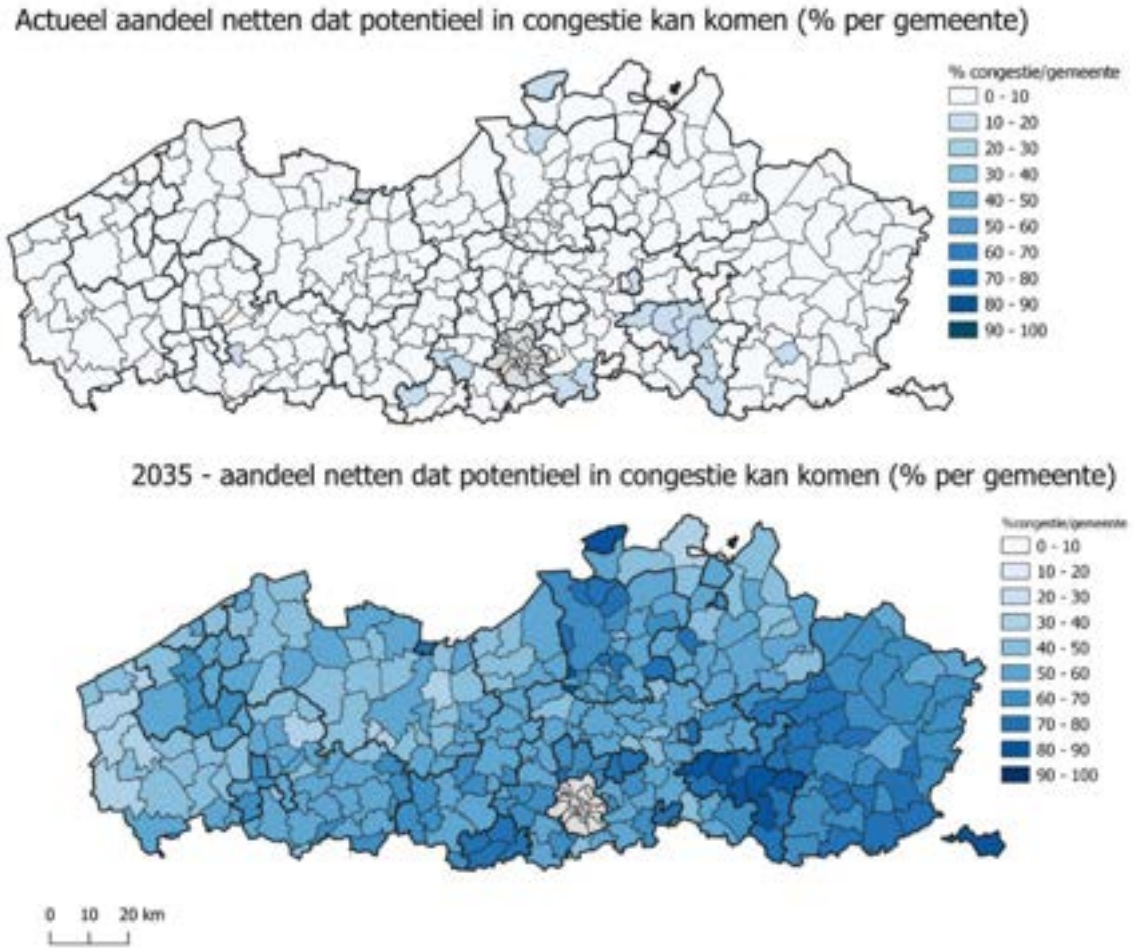
- Note: TSOs and DSOs have complementary options to cope with congestion
  - Grid tariffs (e.g. the capacity tariff)
  - (Flexible) connection agreements
  - Grid investments

- Why do we have congestion?

https://fsr.eui.eu/electricity-markets-in-the-eu/

Redispatch and flexibility markets



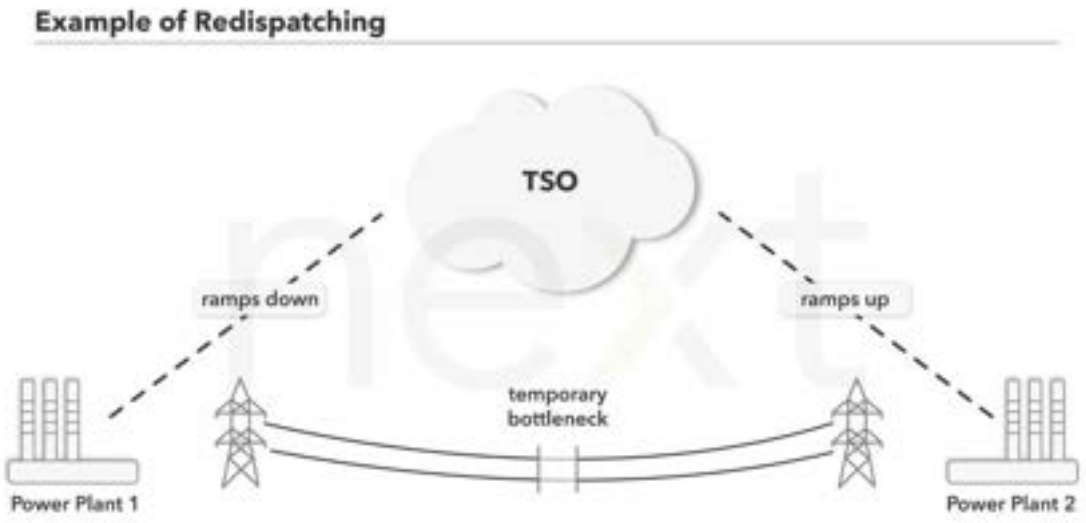


https://www.elia.be/en/customers/connection/grid-hosting-capacity

Fluvius (2023). Investment plan 2023-2032, https://partner.fluvius.be/sites/fluvius/files/2022-06/investeringsplan-2023-2032.pdf

Redispatch markets

- The TSO is responsible for resolving the situation through so-called
  - **Redispatching:** Upwardly and downwardly adjusting individual schedules to meet transmission constraints
  - **Countertrading:** The same principle, but across bidding zones (i.e. multiple TSOs are involved).



https://www.next-kraftwerke.com/knowledge/dispatch

Redispatch markets

- 2 dominant mechanisms:

Figure 1: Illustration of the different scenarios analyzed (top) and our main findings (bottom).

Redispatch regime	CBR		MBR			
Pricing rule for compensation	cost-based pricing		pay-as-bid pricing		uniform nodal pricing	
Objective of the TSO	minimize red. cost	maximize welfare	minimize red. cost	maximize welfare	minimize red. cost	maximize welfare

V. Grimm, A. Martin, C. Sölch, M. Weibelzahl, and G. Zöttl, "Market-Based Redispatch May Result in Inefficient Dispatch," *The Energy Journal*, vol. 43, no. 5, pp. 205–230, Sep. 2022,

- In Belgium: cost-based redispatch where TSO uses balancing energy bids
- Note: market-based preference is the EU-preference, but derogations are possible

Flexibility markets: relatively new

Table 5.1: Examples of the contracted flexibility [MW] and traded flexibility volumes [MWh] on the third-party flexibility market platforms NODES [115], Piclo Flex [118, 119] and GOPACS [121].

	2019	2020	2021	2022
NODES (Norflex) [MWh]	-	-	225.2	734.6
NODES (Norflex) [MW]	-	-	-	3.38
Piclo Flex (UKPN) [MW]	19.3	123.0	350.0	367.6
Piclo Flex (SPEN) [MW]	53.3	139.6	555.0	-
GOPACS (Liander) [MWh]	-	69.2	111.3	7.6
GOPACS (TenneT) [MWh]	36 552.1	53 887.8	142 997.6	181 933.1

Figure 5.1: Projects funded by the European Union's Horizon 2020 research and innovation program focused on developing new flexibility services.

E. Beckstedde, "Fostering Flexibility in Distribution Networks through empirical studies, regulatory analyses, and mathematical programs," PhD dissertation, KU Leuven, Leuven, Belgium, 2023.

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NODES-Fluvius

MaxUsage™

- Fluvius en jouw bedrijf spreken vooraf een **maximaal vermogen** af dat je mag afnemen of injecteren in bepaalde zones.
- Dat plafond geldt tijdens piekmomenten, zodat het net niet overbelast raakt.
- Er is **geen activeringssignaal nodig**: je levert op specifieke momenten, zoals afgesproken in het bod.

Voorbeeld
Een bedrijf in een drukke zone spreekt met Fluvius af dat het verbruik niet hoger gaat dan 2 MW. Zo kan het bedrijf zijn processen plannen binnen die grens, en blijft het net stabiel.

LongFlex™

- Je maakt als bedrijf **op voorhand capaciteit vrij**: een deel van je verbruik of productie houd je flexibel beschikbaar.
- Fluvius betaalt jou een **reservatievergoeding** om dat vermogen klaar te houden.
- Jij biedt het gereserveerde volume aan in de ShortFlex™-markt. Daar koopt Fluvius de activaties aan die het net nodig heeft.

Voorbeeld
Een koelhuis spreekt af dat het elke dag tussen 17 en 20 uur tot 1 MW verbruik kan verminderen als Fluvius dat vraagt. Of dat effectief gebeurt, hangt af van de druk op het net.

ShortFlex™

- Bedrijven bieden dagelijks blokken capaciteit aan. Fluvius kiest welke blokken er die dag effectief worden geactiveerd.
- Er is dus geen vaste reservatie vooraf, maar een **dagelijkse bieding en selectie**.

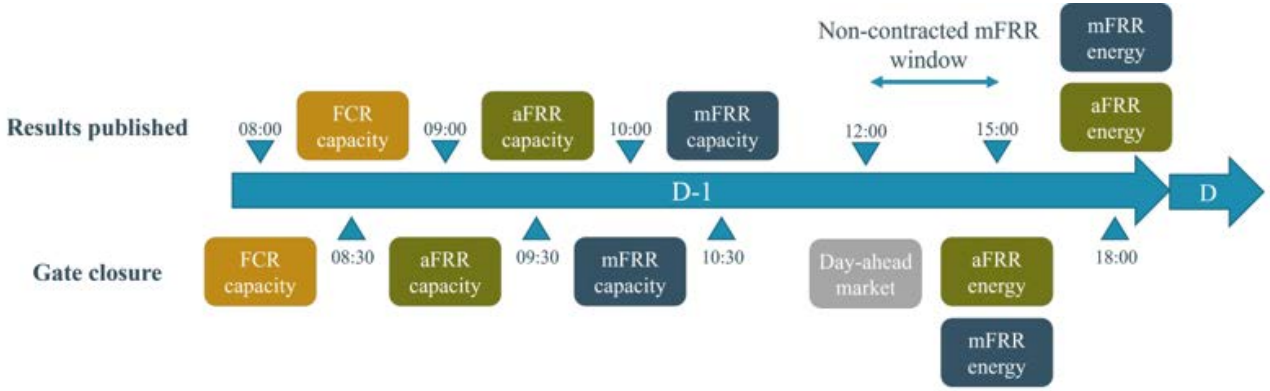
Voorbeeld
Een fabriek biedt op dinsdagmorgen aan dat ze die avond haar compressoren een uur kan stilleggen. Als Fluvius dat bod selecteert, dan zet de fabriek de compressoren uit op het afgesproken tijdstip.

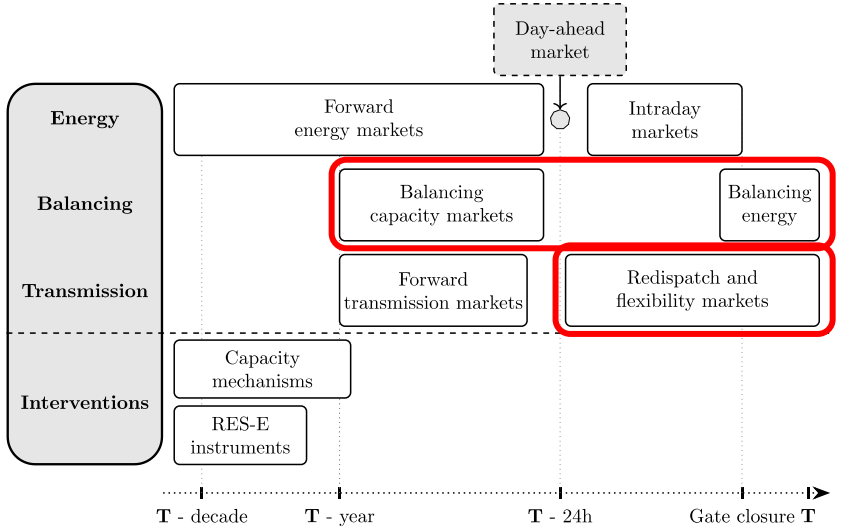
https://partner.fluvius.be/nl/flexibility-service-provider/soorten-marktflex

https://portal.nodesmarket.com/onboarding/tenders?orderBy=openTo&orderByDirection=asc&map-bounds=52.462583147076025,7.053222656250001,50.01820189857675,1.4282226562500002&tab=tenders

Ongoing developments

- Belgium
  - Towards a joint auction for aFRR and mFRR
  - Real-time pricing for improved implicit balancing
  - Go-life of flexibility market Fluvius
- EU
  - Network Code Demand Response
  - PICASSO/MARI participation





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